

Sample Proportions and Normal Approximations — Problem Set

Warm-up

1. A sample of size $n = 100$ is taken from a population with $p = 0.4$. Calculate the mean of the sample proportion $\hat{p}$.
2. Given $p = 0.2$ and $n = 64$, calculate the standard deviation of the sample proportion, $\sigma_{\hat{p}}$.
3. A spinner has 4 equal sectors labeled 1, 2, 3, and 4. What is the value of p for the event "spinning a 2"?
4. If $\hat{p} = \frac{X}{n}$, find $\hat{p}$ when $X = 12$ and $n = 50$.
5. For a binomial distribution to be approximately normal, the conditions $np \geq 10$ and $n(1-p) \geq 10$ are often used. If $p = 0.1$, what is the minimum n required?
6. If Z is a standard normal variable, find $P(Z < 0)$ using the properties of symmetry.
7. A sample of 400 people is surveyed. If the population proportion is 0.75, find the variance of the number of successes, $Var(X)$.
8. If $P(Z < 1.96) = 0.975$, find $P(Z > 1.96)$.
9. A coin is tossed 20 times. If X is the number of heads, state the value of the population proportion p .

Skill-building

1. State the formula for the z -score of a sample proportion $\hat{p}$ in terms of p and n .
2. A machine produces components where 5% are defective. A random sample of 100 components is selected. Calculate the mean and standard deviation of the proportion of defective components.
3. In a large population, 30% of people support a new policy. A sample of $n = 50$ is taken. Find the probability that the sample proportion $\hat{p}$ is exactly 0.2, using the binomial distribution formula.
4. For a population with $p = 0.6$, determine the smallest sample size n such that the standard deviation of $\hat{p}$ is less than 0.05.
5. Using a normal approximation with $p = 0.5$ and $n = 100$, find the z -score corresponding to a sample proportion of $\hat{p} = 0.55$.
6. A fair die is rolled 120 times. Let X be the number of times a '6' is rolled. Find $E(X)$ and $SD(X)$.
7. In a survey of 200 students, the proportion who play an instrument is 0.4. If another student is added to the sample who plays an instrument, what is the new sample proportion?

8. Given $p = 0.8$ and $n = 100$, use the normal approximation to find $P(\hat{p} > 0.84)$. (Assume $P(Z < 1) = 0.8413$).
9. Suppose $\hat{p}$ follows a normal distribution with mean 0.5 and standard deviation 0.02. Find the value of k such that $P(0.5 - k < \hat{p} < 0.5 + k) = 0.95$.
10. A manufacturer claims 90% of bulbs last 1000 hours. In a sample of n bulbs, the standard deviation of $\hat{p}$ is 0.015. Find n .
11. If $X \sim Bi(n, p)$, show that the variance of the sample proportion $\hat{p} = \frac{X}{n}$ is $\frac{p(1-p)}{n}$.

Easier Exam Questions

1. (Blacktown Boys 2024 Q2) A standard six-sided die is rolled 18 times.

Let $\hat{p}$ be the proportion of the rolls with an outcome of 3.

Which of the following expressions is the probability that at most 12 of the rolls have an outcome of 3?

- (A) $P\left(\hat{p} \leq \frac{1}{6}\right)$
- (B) $P\left(\hat{p} \leq \frac{2}{3}\right)$
- (C) $P\left(\hat{p} \geq \frac{1}{6}\right)$
- (D) $P\left(\hat{p} \geq \frac{2}{3}\right)$

2. (Blacktown Boys 2025 Q9) For the population of individuals who own a smart phone, it is known that the proportion of the population that have a particular app is 0.25. A technology company wishes to survey n number of people about this. Let $\hat{p}$ be the random variable representing the proportion of people surveyed that have this app. What is the smallest sample size, n , for which the standard deviation of $\hat{p}$ is less than 0.035?

- (A) 51
- (B) 52
- (C) 153
- (D) 154

3. (Blacktown Boys 2020 Q12a) Bags of lollipops are supposed to contain 50 lollipops. Production records indicate that 92% of bags contain 50 lollipops. A batch of 15 bags is sampled. If more than 2 bags do not contain exactly 50 lollipops, production is stopped.

- i) Find the probability, correct to four decimal places, that exactly 2 of the 15 bags selected do not contain 50 lollipops. **1**
- ii) Find the probability, correct to four decimal places, that production is stopped. **2**

4. (Hunters Hill 2023 Q11e) In a particular country town, the proportion of employment in the farming industry is 0.62. Five people aged 15 years and older are selected at random from the town. Calculate the probability that the proportion of workers in the farming industry in the sample is greater than 0.5, correct your answer to 2 decimal places. **3**

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177

Harder Exam Questions

1. (Glenwood 2023 Q11e) Records show that 64% of students at a school travelled to and from school by bus. A group of 100 students at the school are taken to determine the number of students who travel to and from school by bus.

i) Evaluate $E(X)$ and $\sigma(X)$ 2

ii) Use the table below of $P(Z < z)$, where Z has a standard normal distribution, to estimate the probability that a sample of 100 students will contain at least 58 and at most 64 students who travel to and from school by bus. 2

2. (Sydney Boys 2025 Q12b) It is known that $\frac{3}{5}$ of adult Australians take part in some form of organised sport.

A marketing company is undertaking a survey on consumer preferences for a sports clothing manufacturer.

The marketing company requires a sample of sufficient size. The company needs the probability to be 90% that a member of the sample, who takes part in organised sport, is within 0.025 of $\frac{3}{5}$. The size of the sample is S .

What is the minimum value of S (to the nearest ten people)? 2

You may assume that the sample size is sufficient for the normal approximation to the binomial distribution to be used (without continuity correction).

You are given that if $Z \sim N(0, 1)$ then $P(Z < 1.285) = 0.9$ and $P(Z < 1.645) = 0.95$

3. (Hurlstone 2020 Q12b) A state-wide housing study found that 36% of adults in NSW have a mortgage.

(i) A random sample of 25 adults in NSW is to be taken to determine the proportion of those who have a mortgage. 2

Show that the mean and standard deviation for the distribution of sample proportions of such random samples are 0.36 and 0.096 respectively.

(ii) Part of a table of $P(Z \leq z)$ values, where Z is a standard normal variable, is shown. 2

z	+0.00	+0.01	+0.02	+0.03	+0.04	+0.05	+0.06	+0.07	+0.08	+0.09
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964

Of a random sample of 25 adults in NSW, use the table to estimate the probability that at most three will have a mortgage.

Give your answer correct to four decimal places.

4. (Killara 2020 Q12e) A restaurant knows from past experience that 27% of customers will order a dessert after the main course. For next week, the restaurant has taken 288 customer bookings. Determine the probability that less than 100 will order dessert. A normal distribution table is

included overleaf.

	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0	0.004	0.008	0.012	0.016	0.02	0.024	0.028	0.032	0.036
0.1	0.04	0.044	0.048	0.052	0.056	0.06	0.064	0.068	0.071	0.075
0.2	0.079	0.083	0.087	0.091	0.095	0.099	0.103	0.106	0.11	0.114
0.3	0.118	0.122	0.126	0.129	0.133	0.137	0.141	0.144	0.148	0.152
0.4	0.155	0.159	0.163	0.166	0.17	0.174	0.177	0.181	0.184	0.188
0.5	0.192	0.195	0.199	0.202	0.205	0.209	0.212	0.216	0.219	0.222
0.6	0.226	0.229	0.232	0.236	0.239	0.242	0.245	0.249	0.252	0.255
0.7	0.258	0.261	0.264	0.267	0.27	0.273	0.276	0.279	0.282	0.285
0.8	0.288	0.291	0.294	0.297	0.3	0.302	0.305	0.308	0.311	0.313
0.9	0.316	0.319	0.321	0.324	0.326	0.329	0.332	0.334	0.337	0.339
1	0.341	0.344	0.346	0.349	0.351	0.353	0.355	0.358	0.36	0.362
1.1	0.364	0.367	0.369	0.371	0.373	0.375	0.377	0.379	0.381	0.383
1.2	0.385	0.387	0.389	0.391	0.393	0.394	0.396	0.398	0.4	0.402
1.3	0.403	0.405	0.407	0.408	0.41	0.412	0.413	0.415	0.416	0.418
1.4	0.419	0.421	0.422	0.424	0.425	0.427	0.428	0.429	0.431	0.432
1.5	0.433	0.435	0.436	0.437	0.438	0.439	0.441	0.442	0.443	0.444
1.6	0.445	0.446	0.447	0.448	0.45	0.451	0.452	0.453	0.454	0.455
1.7	0.455	0.456	0.457	0.458	0.459	0.46	0.461	0.462	0.463	0.463
1.8	0.464	0.465	0.466	0.466	0.467	0.468	0.469	0.469	0.47	0.471
1.9	0.471	0.472	0.473	0.473	0.474	0.474	0.475	0.476	0.476	0.477
2	0.477	0.478	0.478	0.479	0.479	0.48	0.48	0.481	0.481	0.482
2.1	0.482	0.483	0.483	0.483	0.484	0.484	0.485	0.485	0.485	0.486
2.2	0.486	0.486	0.487	0.487	0.488	0.488	0.488	0.488	0.489	0.489
2.3	0.489	0.49	0.49	0.49	0.49	0.491	0.491	0.491	0.491	0.492
2.4	0.492	0.492	0.492	0.493	0.493	0.493	0.493	0.493	0.493	0.494
2.5	0.494	0.494	0.494	0.494	0.495	0.495	0.495	0.495	0.495	0.495
2.6	0.495	0.496	0.496	0.496	0.496	0.496	0.496	0.496	0.496	0.496
2.7	0.497	0.497	0.497	0.497	0.497	0.497	0.497	0.497	0.497	0.497
2.8	0.497	0.498	0.498	0.498	0.498	0.498	0.498	0.498	0.498	0.498
2.9	0.498	0.498	0.498	0.498	0.498	0.498	0.499	0.499	0.499	0.499
3	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499

5. (Hurlstone 2023 Q13c) In a certain school, 23% of Year 12 students study Mathematics Extension 1.

- (i) If X represents the number of students who study Mathematics Extension 1, describe the skewness of the binomial distribution for $P(X = x)$.

You must give reasons for your answer.

1

- (ii) The Principal meets with a random sample of 60 Year 12 students, to discuss their Year 12 HSC courses.

What is the probability that more than 30% of the students that meet with the Principal study Mathematics Extension 1?

(You may assume that the sample of students is approximately normally distributed, and make reference to the extract below from a table of z scores).

4

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
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